

1.What is Java? And write the features of Java.

Java is a high-level, object-oriented programming language. It is platform-independent. Features: Simple, Object-Oriented, Platform-Independent, Secure, Multithreaded, Portable, Robust.

2.Why is Java a platform-independent language?

Java is considered a platform-independent language because of its "Write Once, Run Anywhere" (WORA) principle. This is primarily due to the Java Virtual Machine (JVM) and the bytecode compilation process. Here's how it works:

Compilation into Bytecode

When you write a Java program, the Java compiler (javac) converts the source code (.java file) into an intermediate form called bytecode (.class file).

This bytecode is not specific to any operating system or hardware.

Execution by the JVM

The bytecode is executed by the Java Virtual Machine (JVM), which acts as an interpreter.

The JVM is available for different operating systems (Windows, Linux, macOS, etc.).

Since each platform has its own JVM implementation, the same Java bytecode can run on any OS with a compatible JVM.

Abstraction from OS and Hardware

Unlike languages like C or C++, which generate platform-specific machine code, Java's bytecode remains the same across platforms.

The JVM translates this bytecode into machine-specific code at runtime.

3.Can Java be said to be a complete object-oriented programming language?

No, because it supports primitive data types like int, char, float, etc

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4.What do you understand by an instance variable and a local variable?

- Instance Variable: Declared inside a class but outside methods.
- Local Variable: Declared inside a method and accessible only within that method.

5.Explain OOP concepts and write the pillars of OOP with examples.

OOP Concepts: Object, Class, Inheritance, Polymorphism, Encapsulation, Abstraction. Pillars: Encapsulation, Inheritance, Polymorphism, Abstraction.

6.How is an infinite loop declared in Java with examples?

```
while (true) {  
System.out.println("Infinite Loop"); }  
for (;;) { System.out.println("Infinite  
Loop"); }
```

7.Difference between JVM, JRE, and JDK.

- JVM: Runs Java bytecode.
- JRE: Includes JVM and libraries.
- JDK: Includes JRE and development tools.

8.Explain `public static void main(String args[])` in Java.

- `public`: Accessible everywhere.
- `static`: No need to create an object.
- `void`: No return value.
- `main`: Entry point of Java programs.
- `String args[]`: Accepts command-line arguments.

9.Explain different data types in Java.

Primitive: int, char, boolean, float, double, long, short, byte. Non-Primitive: Arrays, Classes, Interfaces, Strings.

10.What are operators and types of operators?

Types: Arithmetic (+, -, *, /, %), Relational (==, !=, >, <), Logical (&&, ||, !), Bitwise (&, |, ^), Assignment (=, +=, -=).

11.Write a Java program to generate the Fibonacci series.

```
public class Fibonacci {  
    public static void  
        main(String[] args) { int  
            n = 10, a = 0, b = 1;  
            for (int i = 1; i <= n; i++)  
                { System.out.print(a + "  
                    ");  
  
                    int sum = a + b;  
                    a = b;
```

```

        b = sum;
    }
}

```

12. Write a Java program for checking prime numbers.

```

public class PrimeCheck {
    public static void
    main(String[] args) { int
    num = 29, flag = 0;
    for (int i = 2; i <= num / 2; i++) {
        if (num % i == 0) { flag = 1; break; }
    }
    System.out.println(num + (flag == 0 ? " is Prime" : " is Not Prime"));
}
}

```

13. Write a Java program for checking palindrome numbers.

```

public class Palindrome {
    public static void
    main(String[] args) {
    int num = 121, rev = 0,
    temp = num; while
    (temp > 0) {
        rev = rev * 10
        + temp % 10;
        temp /= 10;
    }
    System.out.println(num + (num == rev ? " is Palindrome" : " is Not Palindrome"));
}
}

```

14. Write a Java program for checking factorial numbers.

```

public class Factorial {
    public static void
    main(String[] args) { int
    num = 5, fact = 1;
    for (int i = 1; i <=
    num; i++) { fact
    *= i;

```

```
    }  
    System.out.println("Factorial of " + num + " is " + fact);  
  }  
}
```